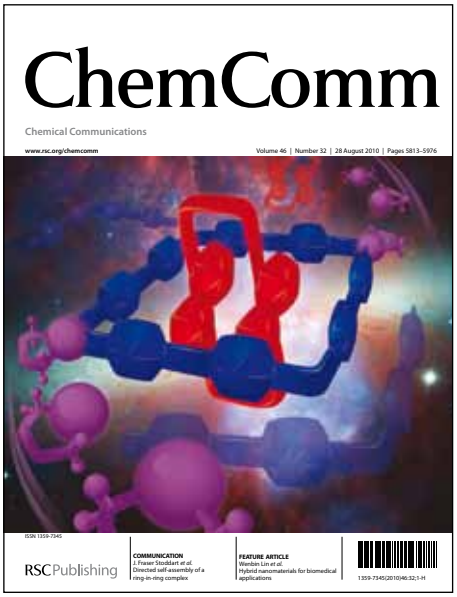
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ARTICLE TYPE

CuBr for KA^2 Reaction: En Route to Propargylic Amines Bearing a Quaternary Carbon Center†

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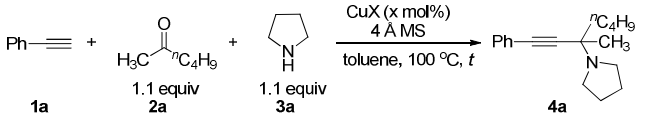
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A highly efficient CuBr-catalyzed coupling of ketones, amines and alkynes (KA^2) forming propargylic amines bearing a quaternary carbon center with so far the broadest scope has been developed.

Propargylic amines are important and useful building blocks in the synthesis of natural products and bioactive compounds.[1] The most straight forward well-established approach is the direct couplings of aldehyde-amine-alkyne (A^3) forming secondary propargylic amines.[2,3] However, the three-component couplings of *ketone*-amine-alkyne (KA^2) have been largely underdeveloped due to the extremely lower reactivity of ketones observed in this reaction.[4] The most commonly used catalysts are copper based:[5-9] For example, Larsen et al. reported only one example that relatively expensive and inconvenient Cu(OTf)$_2$ (10 mol%) could catalyze the reaction of 1-octyne, cyclohexanone and benzylamine to afford the corresponding propargylic amine in 80% yield;[5] $CuCl_2$ has also been used as the catalyst, however, the ketones must be activated by the addition of 50 mol% of Ti(OEt)$_4$ with a very slow reaction rate;[6] Ramón et al. utilized Cu(OH)$_x$-Fe_3O_4 for such a transformation of phenylacetylene with 3-pentanone or acetophenone and hexahydro-pyridine yielding the propargylic amines in very low yields (29-38% yields) taking 7 days.[7] As a comparsion, Cu(I) seems much less active: Van der Eychen et al. applied CuI (20 mol%) to yield the proparglic amine in 30% yield from the single reaction of phenylacetylene, acetone and PMB-NH_2, which should be conducted under the mediation of microwave.[8] Thus, the challenge remains in the extremely narrow scope of the ketones, slow reaction rate, and large amounts of reagents applied. Remarkably, screening of cheap Cu(I) catalysts and solvents in this laboratory revealed that this reaction is indeed particularly rapid and efficient just with CuBr as the catalyst in toluene. Herein, we wish to report a highly efficient CuBr-catalyzed coupling of ketones, amines and alkynes forming propargylic amines bearing a quaternary carbon center with a most broad scope.

Our initial work began with phenylacetylene **1a**, a rather challenging acyclic 2-hexanone **2a**[4] and pyrrolidine **3a** under the catalysis of Cu(I). For the purpose of removing the in-situ formed water in the step of ketiminium formation, 4 Å molecular sieves (MS) were added. After numerous trials, we were happy to find that the reaction of **1a** (1 equiv), **2a** (1.1 equiv), **3a** (1.1 equiv) and CuI (10 mol%) in the presence of 4 Å MS (300 mg·mL^{-1}) in toluene with stirring at 100 °C for 3 h afforded the propargylic amine **4a** in 88% yield (Table 1, entry 1)! Other solvents gave lower yields (Table 1, entries 2-4). When we used CuCl as catalyst, the yield was improved to 98% (Table 1, entry 5)! To our delight, a quantative yield was obtained when CuBr was applied (Table 1, entry 6). Further screening led to the observation that this catalytic protocol is highly efficient: 1.5 mol% of CuBr is enough to produce **4a** in a quantative yield within 1 h (Table 1, entry 9)! Further reducing the reaction time led to a lower yield (Table 1, entry 10). When we changed the ratio of the starting material, the yield dropped (Table 1, entry 11). Thus, **1a** (1 equiv), **2a** (1.1 equiv), **3a** (1.1 equiv), CuBr (1.5 mol%) and 4 Å MS (300 mg·mL^{-1}) in toluene at 100 °C were defined as the optimized reaction conditions for further study.

Table 1 Optimization of Reaction Conditions: the Reaction of Alkyne **1a**, Ketone **2a** and Pyrrolidine **3a** [a]



Entry	CuX	x (mol%)	*t* (h)	Yield of **4a**(%)[f]
1	CuI	10	3	88
2[b]	CuI	10	3	87
3[c]	CuI	10	3	83
4[d]	CuI	10	3	75
5	CuCl	10	3	98
6	CuBr	10	3	100
7	CuBr	1.5	3	100
8	CuBr	0.5	3	92
9	**CuBr**	**1.5**	**1**	**100**
10	CuBr	1.5	0.5	95
11[e]	CuBr	1.5	1	85

[a] The reaction was conducted using alkyne (1.0 mmol), ketone (1.1 mmol), amine (1.1 mmol) and 4 Å MS (300 mg) at 100 °C in 1 mL of toluene. [b] 1 mL of dioxane was used instead of toluene. [c] 1 mL of DMF was used instead of toluene. [d] 1 mL of DMSO was used instead of toluene. [e] Alkyne (1.0 mmol), ketone (1.0 mmol) and amine (1.0 mmol) were used. [f] Determined by ^{1}H NMR analysis using mesitylene as the internal standard.

The scope of the reaction was then investigated by using the optimal reaction conditions (Table 2). Phenylacetylene **1a** and its

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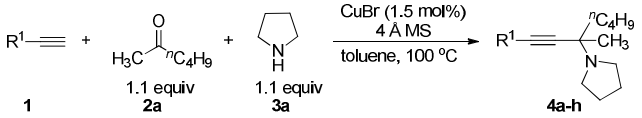
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analogues substituted by *p*-MeO, *p*-Br, *m*-Br, *o*-Cl groups all afforded excellent yields of the products (Table 2, entries 1-5). The reaction could also be extended to terminal alkyl-substituted alkynes such as 1-octyne **1f**, which gave the product in 85% yield (Table 2, entry 6). Moreover, terminal propargylic alcohol **1g** and protected propargylic alcohol **1h** are also suitable for this reaction yielding products **4g** and **4h** in 86% and 78% yields, respectively, showing the functionality-tolerance ability of this protocol (Table 2, entries 7 and 8). The hydroxyl group and TBS protected hydroxyl group of the propargylic amines **4g** and **4h** could be easily converted into other functional groups, which demonstrate the potential of this method.

Table 2 The Scope of Terminal Alkynes[a]

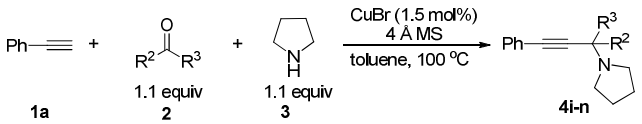


Entry	R^1 (**1**)	*t* (h)	Isolated Yield of **4** (%)
1	Ph (**1a**)	1	92 (**4a**)
2	4-$MeOC_6H_4$ (**1b**)	3.5	86 (**4b**)
3	4-BrC_6H_4 (**1c**)	3	87 (**4c**)
4	3-BrC_6H_4 (**1d**)	1.5	90 (**4d**)
5	2-ClC_6H_4 (**1e**)	1	92 (**4e**)
6	*n*-Hex (**1f**)	4	85 (**4f**)
7	$HO(Me)_2CH$ (**1g**)	2	86 (**4g**)
8	$TBSOCH_2$ (**1h**)	4	78 (**4h**)

[a] The reaction was conducted using alkynes **1** (10 mmol), 2-hexanone **2a** (11 mmol), pyrrolidine **3a** (11 mmol), CuBr (1.5 mol%) and 4 Å MS (3 g) at 100 °C in 10 mL of toluene. TBS = *tert*-butyldimethylsilyl.

Encouraged by these results we further investigated a range of different ketones with phenylacetylene **1a** as the standard alkyne. The reaction showed a very broad substrate scope: not surprisingly, methyl ketones, such as 2-butanone **2b**, 2-octanone **2c**, benzylacetone **2d** and 3-methyl-2-butanone **2e** all afforded propargylic amines **4i-4l** in 80~89% yields (Table 3, entries 1-4); the reaction of ethyl ketone also worked (Table 3, entry 5); even with heptane-4-one **2g**, the yield of propargylic amine **4n** is 66% although 5% of CuBr was applied (Table 3, entry 6); when cyclohexanone **2h** was used, the corresponding product **4o** was yielded in 93% (Table 3, entry 7).

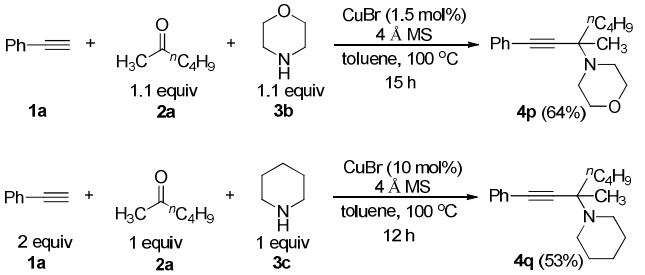
Table 3 The Scope of Unactivated Ketones and Amines[a]



Entry	R^2, R^3	*t* (h)	Isolated Yield of **4** (%)
1	Me, Et (**2b**)	2	87 (**4i**)
2	Me, *n*-C_6H_{13} (**2c**)	1.5	86 (**4j**)
3	Me, CH_2CH_2Ph (**2d**)	2	89 (**4k**)
4	Me, *i*-Pr (**2e**)	2.5	80 (**4l**)
5	Et, $(CH_2)_2CH_3$ (**2f**)	2.5	84 (**4m**)
6[b]	$CH_3(CH_2)_2$, $(CH_2)_2CH_3$ (**2g**)	8	66 (**4n**)
7	-$(CH_2)_5$- (**2h**)	3	93 (**4o**)

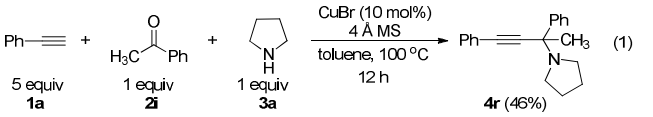
[a] The reaction was conducted using alkyne **1a** (10 mmol), ketones **2** (11 mmol), pyrrolidine **3a** (11 mmol), CuBr (1.5 mol%) and 4 Å MS (3 g) at 100 °C in 10 mL of toluene. [b] CuBr (5 mol%) was used.

The reaction may also be extended to other cyclic amines such as morphine and piperidine with relatively lower yields (Scheme 1). Other acyclic amines, such as benzylamine and dibenzylamine, do not work under the current reaction condition, indicating the effect of amine on this protocol.

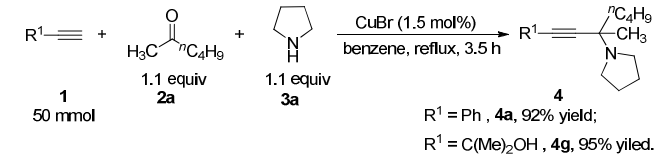


Scheme 1 Cyclic Amines in KA[2] Reaction.

It is worthwhile to note that aromatic ketones, which are totally inactive in the recently reported methods,[6] also work in this protocol: with 10 mol% of CuBr the corresponding product **4r** was formed in 46% yield (Eq. 1).[7]



It is easy to conduct the reaction on 50 mmol scale in benzene to afford **4a** and **4g** in 92% and 95% yields just using a Dean-Stark trap to remove the in-situ formed water in the step of ketiminium formation (Scheme 2), thus, the addition of the molecular sieves is not necessary in any large scale practical application.



Scheme 2 Approach to Propargylic Amine (**4**) on 50 mmol Scale Using a Dean-Stark Trap.

Conclusions

In summary, we have developed a highly efficient CuBr-catalyzed direct coupling of unactivated ketones, amines and alkynes to form propargylic amines bearing a quaternary carbon center. The mild reaction conditions, the broad scope of the substrates and the high efficiency of the reaction show the potentially synthetic utility of this method.[10-12] Further studies including the asymmetric studies and the application of the propargylic amines are being conducted in our laboratory.

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Notes and references

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† Electronic supplementary information (ESI) available: The experimental procedures, characterization data, and copies of ^{1}H and ^{13}C NMR spectra for all products. For ESI or other electronic format see DOI:

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